Polycheles typhlops, Heller.

Pl. I, figs. 1–13.

Polycheles typhlops, Heller, 1862.
Polycheles typhlops, Heller, 1863.
Polycheles typhlops, Carus, 1885.
Polycheles Doderleini, Riggio, 1885.
Pentacheles Hextii, Alcock, 1894.
Polycheles typhlops, Adensamer, 1898.
Pentacheles Hextii, Alcock, 1901.
Polycheles typhlops, Senna, 1903.
Polycheles typhlops, Bouvier, 1905 (a).
Polycheles typhlops, Bouvier, 1905 (b).
Polycheles typhlops, Kemp, 1912.

The carapace is slightly shorter than the abdomen. Its dorsal surface is almost flat, and its lateral margins are very slightly convex. The frontal margin considered as a whole is slightly concave, and is divided into three lobes by the deep sinus at each side, in which the fixed and immovable eyestalk lies. The median lobe is bounded at either side by a strong spine. In the centre is the rostrum, formed by a single sharp spine pointing forwards and upwards. Below this there is a broader and blunter process in the median line. The outer border of each orbital sinus is furnished with three or four small teeth, which are more distinct in the female than in the male. At each antero-lateral angle of the carapace there is a strong curved spine, and from this there runs backwards a line of teeth along the lateral margin. These lateral spines decrease in size towards the posterior end, and the series is divided into three parts by the two branches of the cervical groove. The front part has usually seven spines, the middle four or five, and the hind part about twenty. The numbers however vary considerably. Senna gives them as 6 or 7, 4, 14 to 18, and Carus 7, 5, 20 to 23. The two sides of the same specimen even may show differences in the number of spines; in one of the specimens taken by the *Helga* the left side has 7, 5, 18, and the right, 8, 5, 20. The hind carapace margin is strongly concave, and is composed of a strong bar, the two halves of which meet at an obtuse angle in the centre. At this point there are two large forward-pointing spines, and two smaller ones further down on either side. The posterior extremities of this bar are firmly buckled to the first abdominal segment. From the middle of the hind margin to the rostrum there runs a raised carina along the whole length of the carapace. It is broken only at one point, where it is crossed by the cervical groove. It bears several distinct spines, some of which are double, and between these it is

covered with rough tubercles. In front of the cervical groove the spines, according to Senna, are arranged as follows

Rostrum + 1, 1, 1, 1, 2, 2,

in the male, and in the female either the same or,

Rostrum + 1, 1, 1, 2, 2.

In the specimens taken by the *Helga* these spines are not always well developed. It is especially difficult sometimes to say whether a spine is double or not. The one which lies just in front of the cervical groove is very often single, or has one half much larger than the other, or is altogether rudimentary. Behind the groove there are three double spines, but often some of these are very slightly developed. The arrangement of the carinal spines on some of the Irish specimens is as follows. (The semicolon marks the position of the cervical groove, and *r* signifies that the spine is rudimentary.)

Rostrum + 1, 1, 1, 2, 1½ ; 2, *r*, 2.
Rostrum + 1, 1, 1, 2, *r* ; *r*, *r*, 2.
Rostrum + 1, 1, 1, 1, 2, 1½ ; *r*, *r*, 2.
Rostrum + 1, 1, 1, 2, 1½ ; 2, 2, 2.
Rostrum + 1, 1, 1, 2, *r* ; *r*, *r*, 2.

The cervical groove is situated almost exactly half-way between the rostrum and the hind margin of the carapace. It is well marked, and at a point rather nearer the median line than the margin it divides into two. The posterior branch cuts the margin almost at right angles, whereas the anterior, which is the fainter of the two, slants forwards, and meets the margin at an acute angle. From the hind branch of the cervical groove a low ridge furnished with about twelve to fourteen small teeth runs backwards parallel with the margin to the posterior end of the carapace. It is almost twice as far from the median carina as from the lateral margin. From the middle of the anterior branch of the cervical groove to the orbital sinus there is a curved row of four or five spines, which are not situated on a ridge. On the cervical groove itself, besides the two central spines forming part of the median carina, there is a single smaller spine on each side above the bifurcation, and three or four very small ones on the hind branch below it. Besides the larger spines, the position of which has been indicated, there are numerous very small ones scattered over the surface of the carapace; they are particularly plentiful about the borders of the cardiac region, and all of them are directed forwards.

On the lower surface of the carapace there are at each side two oblique carinae armed with small teeth. The external or upper of these submarginal ridges begins just opposite the base of the antenna and runs backwards towards the postero-lateral angle of the carapace. Its anterior end is marked by a large spine. The teeth are fairly large in the front portion, but they

die away about the middle, and the hind part is only faintly traceable. The internal submarginal carina runs from the posterior angle of the carapace forwards to a point opposite the base of the first pereiopods. It is more definitely spiny than the other.

The tergum of the first abdominal somite is almost quite smooth, but on the second there is a well developed median spine, and on the third, fourth, and fifth there are large spines curving forwards. The sixth segment bears a low double keel which is broken up into six or seven tubercles.

The pleura of the first segment are the smallest, and those of the second the largest of the series. The others diminish in size and increase in sharpness from the third to the sixth.

The basal part of the telson has a median row of three or four tubercles. It tapers to a narrow but not very sharp point.

The pigmentless and immovable eyestalks fill the deep orbits in the front of the carapace. Each one gives off a tapering branch on the outer side, which extends beneath the antero-lateral angle of the carapace. There is a short spine on the front of each eyestalk.

The inner flagella of the antennules are longer than the carapace; the outer are very short and much thinner than the others. The antennular scale extends far past the end of the peduncle, and ends in a sharp point; its inner margin is armed with teeth. The outer angle of the basal joint of the peduncle bears two prominent spines.

The narrow antennal scale does not reach the end of the peduncle. The flagellum is slightly shorter than the inner branch of the antennule. From the basal joint of the peduncle there projects on the inner side a short rounded process, the phymacerite or renal tubercle, the end of which lies in a hollow on the under side of the base of the antennule.

The cutting edge of the mandible bears about a dozen teeth, of which one at each end and one in the middle are larger than the others. There is a three-jointed palp.

The two lobes of the first maxillae are strongly incurved; they are very slender, and end in sharp bristles, which are stouter in the anterior one.

In the second maxillae the two lobes are not subdivided, but are long and slender, the inner larger than the outer; the scaphognathite is very large.

The first maxillipedes are very long and of unusual form. Fig. 7 shows the appearance of the anterior part without the epipodal lamella, which forms a prolongation backwards, so that the total length of the appendage is greater than that of the scaphognathite. The two lobes of the protopodite are represented by the parts *a* and *b*. From *b* there springs the long, slender endopodite, heavily fringed with setae, and having an imperfect joint slightly nearer the tip than the base. The exopodite, on the other hand, is very broad; in its distal part it divides into two curved plates which are curiously curled round

one another. That which is seen in profile in Fig. 7, *d*, partially encloses the other, *e*, which spreads out into a broad lamella from a very narrow base at *e*. On the anterior edge of the proximal protopodial lobe there are two rows of setae, one directed outwards, the other inwards: the former are plumose, the latter simple. The lobe *a* appears to be narrower in the figure than it really is, owing to foreshortening.

Neither the second nor the third maxillipedes have an exopodite. Both appendages are slender and pediform. In the second maxillipedes the carpus is long and much expanded; the dactyl ends in a spine. The third maxillipedes are very long and slender, and are unarmed except for the spines on the dactyl. The whole appendage is covered with long setae. There is a small epipodite.

In the female all five pairs of pereiopods are chelate, but the last pair are not perfectly so, the dactyl being slightly longer than the projecting part of the propodite. In the male the first four pairs are chelate, but the fifth ends simply. In both sexes the tips of the fingers in the first and second pereiopods cross one another and are strongly curved.

The first pereiopods or chelipeds are slightly longer than the whole body from rostrum to tip of telson. They are remarkably slender and feeble. The ischium is very narrow at the proximal end, but expands distally, and this is also true of the carpus.

The merus is broad at the proximal end, narrows in the middle, and then widens again. The palm of the propodite is the stoutest part of the whole appendage, and is not so compressed as the ischium and merus. The fingers are very long and thin, and are sharply bent near the tip. The inner edge of each finger is furnished with a row of small plates set obliquely, and overlapping one another. The merus bears one or two spines at its upper distal end, and the carpus also has one in the same position. Sometimes the propodite bears a spine opposite the base of the dactyl. The upper margin of the merus is very minutely spinulose.

The other pereiopods are very much shorter than the first, and diminish in size from the second to the fifth.

The first pair of pleopods are very different in the two sexes. Those of the female are fairly large and robust, passing the base of the fourth pereiopods, and are heavily fringed with setae, which are of two kinds. There is a continuous fringe of slender pinnate setae along each margin, and in addition to these there is a much smaller number of simple hairs, which are much longer and thicker than the pinnate ones. The largest of all are to be found in a little group of four or five situated at the base of the distal joint.

The first pleopods in the male are quite devoid of hairs, with the exception of five or six on the proximal joint, and one at the extreme tip. Near the extremity there is a slight swelling on the inner side, which bears a group of little hooks. The appen-

dage is very slender, and is considerably shorter than in the female.

The succeeding pairs of pleopods are all well developed, and possess an *appendix interna*, and in the male the second pair have also an *appendix masculina*.

The uropods are large and broad, the exopodite more so than the endopodite. The margins are entire and fringed with long setae. In the endopodite there is a midrib extending to the tip, and in the exopodite two similar ribs, neither of which reach the margin.

Occasionally in the female there may be seen two spermatophores cemented firmly to the sternal surface between the coxae of the fourth and fifth pereiopods. These are not unlike the spermatheca which is found in certain Decapods in the same position, and in *P. sculptus* they have erroneously been described as such (see Andrews, 1911).

Size :—Senna (1903) mentions three large females, taken in the Mediterranean, which measured 65, 70, and 90 mm. The largest specimens taken by the *Helga* are two females, each 80 mm. long. Another female measures 74 mm.

The following measurements are taken from a female from station S. R. 490.

Total length	..	80 mm.
Length of carapace	..	35 mm.
Length of abdomen	..	45 mm.
Breadth of carapace	..	23 mm.
Length of chelipeds	..	81 mm.
Length of ischium of chelipeds	..	13 mm.
Length of merus of chelipeds	..	26·5 mm.
Length of carpus of chelipeds	..	13·5 mm.
Length of propodite of chelipeds	..	27·5 mm.
Length of palm of chelipeds	..	12 mm.
Length of fingers of chelipeds	..	15·5 mm.
Length of second pereiopods	..	32 mm.
Length of third pereiopods	..	28 mm.
Length of fourth pereiopods	..	26 mm.
Length of fifth pereiopods	..	22 mm.
Length of antennules	..	43 mm.
Length of antennae	..	Broken.

General Distribution :—The species was taken originally in the Mediterranean, and was described by Heller (1862 and 1863). It has since been recorded from Sicily (Heller, Riggio), Sardinia (Senna), the Adriatic, north coast of Africa, south coast of Asia Minor, and Crete (Adensamer). Under the name of *P. Agassizi* it has been recorded from several localities in West Indian waters (Milne Edwards), and Bouvier has shown that it is widely distributed in the eastern Atlantic. Kemp (1912)

has shown that Alcock's species *Pentacheles Hextii* is a synonym of *P. typhlops*, so that the range of the species is extended to the Indian Ocean.

Norman (1879) gives a description of two specimens of *Polycheles* taken by the *Porcupine*, one, a male, off the Portuguese coast, and the other, a female, on the Atlantic slope west of the English Channel (48° 13′ N., 9° 11′ W.). The male appears from the description to belong to *P. typhlops*, but the female certainly does not.*

Vertical Distribution :—Riggio records a Mediterranean specimen from about 55 fms., but usually the depths are very much greater. Senna's specimens were taken in 358–848 fms., and Adensamer's in 339–1122 fms. The depths at which the *Helga* has taken this species range from 208 to 728 fms.

Irish Distribution :—The species has not previously been recorded from British waters. All the specimens were found off the south-west coast.

Helga :—

S. R. 171—5 XI '04.—52° 7′ N., 11° 58′ W., 337 fms. Trawl.—Two, 63–34 mm.

S. R. 188—3 II '05.—51° 53′ N., 11° 59′ W., 320–372 fms., mud. Trawl. Temperature at 300 fms., 10·125° C., Salinity 35·5 °/₀₀.—One, 70 mm.

S. R. 321—1 V '06.—50° 56′–51° 0′ 30″ N., 11° 17′ W., 480–208 fms., fine sand. Trawl.—One, 74 mm.

S. R. 353—6 VIII '06.—50° 37′–50° 40′ N., 11° 32′ W., 250–542 fms., mud and sand. Trawl.—Temperature at 500 fms., 8·58° C., Salinity 35·46 °/₀₀.—Seven, 80–30 mm.

S. R. 482—29 VIII '07.—51° 6′ N., 11° 26′ W., 368 fms., fine sand. Trawl.—Three, 55–32 mm.

S. R. 490—7 IX '07.—51° 57′ 30″ N., 12° 7′ W., 470–491 fms., ooze. Trawl. Temperature at depth 6·68° C.—One, 80 mm.

S. R. 502—11 IX '07.—50° 46′ N., 11° 21′ W., 447–515 fms. Trawl. Temperature at depth 8·8° C., Salinity 35·37 °/₀₀.—Three, 50–30 mm.

S. R. 504—12 IX '07.—50° 42′ N., 11° 18′ W., 627–728 fms., coral. Trawl.—One, 32 mm.

S. R. 1242—14 VIII '11.—51° 27′ N., 11° 55′ W., 550–590 fms. Trawl.—One, 52 mm.

S. R. 1693—20 VIII '13.—51° 30′ N., 11° 51′ W., 500–479 fms. Trawl.—One, 76 mm.

* Dr. Calman informs me that the female specimen has been seen by Mr. Kemp, and was referred by him to *P. sculptus*.